



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SESSION 2025 / 2026 SEMESTER 2
SECV2223 WEB PROGRAMMING
REPORT

LECTURER : DR. TS. NOR AZMAN BIN
ISMAIL
SECTION 05

GROUP MEMBERS

NO.	NAME	MATRIC NO.
1	YOUSRA HATIM ELZABIR ALI	A23CS4059
2	NISHAT TABASSUM	A24CS4101
3	Zaky Zulhadi	A24CS4097
4	Fadhil Atha Ramadhan	A24CS4093

Assignment 2 & 3 Combined Short Report

1. Introduction

This short report summarizes the development of the Web Programming project for Assignment 2 and Assignment 3. The system is an academic portfolio and news management website for Prof. Ts. Dr. Nor Azman Ismail. Assignment 2 converted the previous frontend prototype into a full-stack CRUD application using PHP and MySQL, while Assignment 3 extended the same system by integrating a real Gemini AI API for intelligent writing assistance. The final system allows users to manage news records, store them in a database, display them on the public News & Activities page, and use AI features inside the browser through a secure PHP backend.

2. Project Overview

The website contains public academic pages such as Home, About, Research, Publications, Teaching, Contact, and News & Activities. The administrative page, newsEdit.html, works as a news editor where the group can create, view, update, and delete news items. The public page, news.html, fetches saved news from the backend API and displays them with category filters and a Read More option. The project was tested locally using XAMPP, Apache, MySQL, phpMyAdmin, HTML, CSS, JavaScript, PHP, and MySQL.

3. Assignment 2: Backend CRUD Implementation

Assignment 2 introduced a PHP REST API named api.php and a MySQL database named news_editor. The main table, documents, stores the news title, summary content, full details, category, display date, image URL, created timestamp, and updated timestamp. The frontend communicates with the backend using JavaScript fetch() requests. This replaced the previous localStorage-based prototype with real server-side persistence.

Method	Endpoint	Purpose
GET	api.php	Retrieve all news records
GET	api.php?id=1	Retrieve one selected news record
POST	api.php	Create a new news record
PUT	api.php?id=1	Update an existing news record
DELETE	api.php?id=1	Delete a news record

The CRUD operations were verified from the editor page and the database. Create News inserts a new record, Save updates the selected record, Delete removes a record after confirmation, and Read retrieves all records into the sidebar. The public News & Activities page also reflects the updated data from MySQL.

4. Database and Frontend Flow

- config.php stores the local database connection settings for XAMPP and MySQL.
- api.php receives JSON requests and returns JSON responses for all CRUD operations.
- writeMain.js contains DocumentService for API communication and UIController for editor actions.
- main.js loads saved news into the public News & Activities timeline and supports category filtering.
- phpMyAdmin was used to inspect the documents table and export the updated SQL dump.

5. Assignment 3: Gemini AI Integration Assignment 3 required replacing mock AI tools with real AI-powered features. The selected AI provider is Google Gemini because it supports text generation suitable for writing improvement and summarization. To keep the API key secure, the frontend does not call Gemini directly. Instead, the browser sends a request to ai_handler.php, and the PHP backend sends the prompt to Gemini. The real API key is stored only in a private .env file, while .env.example is included in the submission with a placeholder key.

AI Feature	Frontend Trigger	Backend Endpoint	Output
Improve Writing	Improve Writing button	POST ai_handler.php?action=improve	Professional academic wording for the Details text
Summarize Content	Summarize button	POST ai_handler.php?action=summarize	Short summary inserted into Summary / Excerpt

6. AI Feature Design and Testing

The Improve Writing feature sends the selected news title, category, and Details text to Gemini with a prompt instructing it to rewrite the paragraph professionally for a university professor's academic portfolio. The prompt also tells Gemini not to invent new facts and to preserve names, dates, numbers, places, awards, and project details. The Summarize Content feature sends long Details text to Gemini and requests a concise two-sentence summary suitable for the News & Activities page. The summary is placed directly into the Summary / Excerpt field and can be saved to the database.

Test Area	Result
Improve Writing	Successfully returned improved academic wording through Gemini
Summarize Content	Successfully generated a short summary and inserted it into Summary / Excerpt
CRUD after AI	Create, Read, Update, Delete, image URL, and public news display still worked
Security	The Gemini API key was not exposed in JavaScript or .env.example

7. Challenges, Reflection, and Conclusion

The main challenges were connecting Gemini through PHP, fixing model and API request errors, and ensuring that adding AI did not break the existing CRUD functions. These issues were solved by testing `ai_handler.php` separately in the browser console, using the correct Gemini model, and carefully replacing the mock AI service inside `writeMain.js`. The integration improves the user experience because administrators can quickly convert rough notes into professional academic news and produce concise summaries for public display. Overall, Assignment 2 and Assignment 3 were completed as one integrated system: a full-stack CRUD news editor enhanced with real Gemini AI features.

8. Presentation

- YouTube video link: <https://youtu.be/Qu8Qtech1WQ>